Michael M. Fawehinmi

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PROFESSIONAL SUMMARY

Senior Data Engineer with 10+ years in big data analytics, combining cloud data architecture, AI, and data engineering: led a Workday-to-Snowflake integration as sole engineer, unified four workforce platforms into Snowflake, designed raw/staging/production layers with row access policies and dynamic masking, built Snowflake Cortex LLM agents for ~130 C-suite and HRBP users and prototyped on Amazon Bedrock.

- Builds end-to-end data and AI pipelines in Python, SQL, and Snowflake, spanning ETL architecture, schema certification, and LLM integration.
- Built LLM applications and multi-agent orchestration on Snowflake Cortex, including AI-powered Power BI assistants for C-suite leaders and HRBPs, eliminating all 50+ recurring Tier 0 and Tier 1 reporting requests.
- Hands-on agentic AI and deep learning: Snowflake-managed MCP servers exposing 55+ tools to LLM agents, and CNN transfer-learning models (DenseNet121, Xception) for AI-generated image detection.
- 10+ years in big data analytics and data engineering, with deep specialization in AI, Machine Learning, and Snowflake Cortex LLM development.
- Delivers executive-facing analytics and self-service reporting that surface actionable insights on demand.

PROFESSIONAL EXPERIENCE

NetApp, Waltham MA (remote) January 2026 – Present

Senior Data Engineer *(contract – 6 months)*

- Engineered a SAP SuccessFactors (OData API) to Snowflake integration for a 12,000-employee workforce, with ETL pipelines, stored procedures, and scheduled migration tasks.
- Unified four workforce platforms (SuccessFactors, Fieldglass, Eightfold, Visier) into Snowflake and One Model via SFTP, API, and custom connectors.
- Built Snowflake Cortex LLM agents for ~130 C-suite and HRBP users, managing and orchestrating a master agent that delegates to specialized subagents exposed as MCP tools.
- Authored complex SQL views across certified schema layers, keeping multi-process ETL workflows and downstream reporting consistent.
- Architected Power BI service-account schemas with DirectQuery against Snowflake for scalable, performant enterprise reporting.
- Built Streamlit applications for real-time pipeline monitoring, proxy connectivity management, and on-demand data refresh.
- Produced back-to-office, global giving, and compensation reference reporting for executives, HRBPs, and people leaders across 12,000 employees.
- Modeled employee commute distance and time with the Google Maps API and Python traffic models to inform back-to-office policy.

Novanta Corporation, Bedford MA (remote) January 2021 – December 2025

Senior Data Engineer *(full-time)*

- Spearheaded the Workday-to-Snowflake integration as sole engineer, from planning and requirements through architecture and production deployment.
- Integrated Workday data for 3,500 employees into Snowflake via Web Services and Public API, designing raw/staging/production layers, row access policies, dynamic data masking, and nightly maintenance procedures.
- Connected SharePoint to the enterprise data platform using Microsoft Fabric and Snowflake Open Catalog.
- Built LLM applications and agents on Snowflake Cortex (Cortex Analyst, Cortex Search) for ~15 C-suite and HRBP users, encoding years of domain logic, business rules, and conditional logic into AI schemas.
- Prototyped LLM-powered reporting assistants with Amazon Bedrock alongside the Snowflake Cortex implementation.
- Eliminated all 50+ recurring Tier 0 and Tier 1 reporting requests by replacing manual query workflows with self-service reporting built on Snowflake Cortex AI.
- Created a Python toolset of statistical, algorithmic, and ML models (scikit-learn, PyTorch; training on Amazon SageMaker) supporting data-driven decision-making across the organization.
- Delivered a library of 50+ Power BI dashboards for the C-suite and CHRO with an interactive, user-friendly executive interface.
- Quantified the monetary impact of human capital movement with workforce planning and financial models.
- Launched a business-level sensitivity analysis platform for monitoring capacity planning and human capital demand.

Salesforce, Boston, MA (remote) July 2020 – December 2020

Lead Python Developer *(contract)*

- Defined integration architecture and solution requirements, translating business rules into precise data validation and processing logic.
- Implemented Workday Web Services and REST API pipelines for enterprise-scale workforce data migration (ingestion, transformation, load).
- Led design and delivery of a full-scale Python data conversion application for SAP SuccessFactors with validation and error handling.
- Instituted automated data quality assurance with actionable failure reporting, ensuring reliable system-to-system data integrity.
- Tracked pipeline reliability and throughput across the full conversion lifecycle with performance monitoring dashboards.

Global Atlantic Financial Group, Brighton, MA (remote) December 2017 – June 2020

Shared Services Data Engineer *(full-time)*

- Solely designed and built the Greenhouse integration via REST API endpoints connecting Oracle HR systems; maintained and enhanced the existing SuccessFactors integration.
- Assessed vendor platforms and solution architectures for integration fit, compatibility, and infrastructure requirements.
- Owned the end-to-end ML model lifecycle (versioning, monitoring, drift analysis) for GLM, GLMM, and SARIMAX attrition and growth models.
- Automated ingestion, processing, and reporting pipelines for Bureau of Labor Statistics and Johns Hopkins COVID-19 data.
- Surfaced predictive model outputs and workforce planning insights for the C-suite and CHRO through executive dashboards.

- Kept HR, IT, and external vendors aligned on integration milestones as technical liaison, running biweekly cross-functional delivery meetings.

SELECTED AI PROJECTS

Household Data & Agent Platform, personal project 2026

Snowflake-Managed MCP Servers for LLM Agents *(Snowflake, Cortex, SQL, Python)*

- Designed and deployed two Snowflake-managed MCP servers exposing 55+ tools to LLM agent clients (Cursor, Claude, Grok): Cortex Agents, 15 Cortex Analyst semantic views, Cortex Search services, and JSON-schema-typed stored-procedure tools.
- Built the underlying warehouse (RAW/CORE/ANALYTICS/CTRL layers, star schema) with Gmail, Google Drive, and GraphQL API ingestion via external access integrations, AI_CLASSIFY email triage, document OCR extraction, and scheduled tasks.
- Applied agent guardrails: least-privilege MCP client roles, a separate read-only endpoint that excludes write tools, and an async job queue with status polling to keep long-running tools within MCP client timeouts.
- Built a second MCP server over a YouTube content catalog: transcript embeddings (snowflake-arctic-embed-m, 768-d) with vector semantic search, Cortex Search, and a 6-hourly YouTube Data API sync task.

Worcester Polytechnic Institute, graduate ML course project (CS539) Summer 2023

AI-Generated and Inpainted Image Detection with CNNs *(TensorFlow/Keras)*

- Built a detector for AI-generated and inpainted images using CNN transfer learning: ImageNet-pretrained DenseNet121 and Xception backbones fine-tuned in TensorFlow/Keras on the CIFAKE (120K images) and DeepfakeArt inpainting benchmarks.
- Stacked both CNNs into a learned ensemble; achieved held-out test accuracy of 96.4% (CIFAKE), 81.1% (DeepfakeArt), and 94.9% (ensemble).
- Trained with data augmentation, dropout and L2 regularization, early stopping, learning-rate scheduling, and hyperopt tuning, distributed across two GPUs with tf.distribute MirroredStrategy and tracked in TensorBoard.

EDUCATION

Worcester Polytechnic Institute, Worcester, MA
Master's in Data Science | Pioneering Program, Attended
Relevant Coursework: Data Analytics and Mining, Statistical Methods for Data Science, Deep Learning and Machine Learning, Database Management Systems, Bayesian Statistics, Applied Multivariate Statistics

Bentley University, Waltham, MA
Bachelor of Science in Finance with Concentration in Capital Markets, Attended
Relevant Coursework: Quantitative Portfolio Management, Advanced Probability and Statistics, Advanced Financial Modeling, Advanced Concepts in Programmatic Modeling

SKILLS & CERTIFICATES

Data Engineering: Snowflake (Snowpark, Row Access Policies, Dynamic Masking, Stored Procedures, Tasks, Open Catalog, External Access Integrations); ETL/ELT Pipeline Design; Schema Certification; SQL; Spark; Hadoop; Alteryx; Control-M; FTP/SFTP; APIs (REST, OData, SFTP); Web Scraping; Data Mining
Cloud & BI: Microsoft Azure; Azure Data Factory; Microsoft Fabric; AWS S3; Amazon SageMaker; Amazon Bedrock; Power BI (Direct Query, Service Accounts, DAX); Tableau; Streamlit
AI / ML: Snowflake Cortex (LLM, Cortex Analyst, Cortex Search, Cortex Agents); Model Context Protocol (MCP) servers; LLM Agent Tooling; Embeddings & Vector Search; CNNs & Transfer Learning (DenseNet121, Xception); TensorFlow/Keras; PyTorch; Scikit-learn; Statsmodels; SciPy; Hyperopt; Machine Learning; Deep Learning; Bayesian Statistics; Gaussian Process; GLM/GLMM/SARIMAX
Languages & Tools: Python (Pandas, NumPy, Matplotlib); R; C#; Git/GitHub; Agile/Waterfall; Microsoft Office; Workfront
HR & Enterprise Systems: Workday (Web Services, REST API, Public API); SAP SuccessFactors (OData API); Fieldglass; Eightfold; Visier; Greenhouse; One Model; Microsoft Dynamics; Google Maps API; Kiteworks; Beacon
Certificates: Data Analytics Certificate (x3); Database Architectural Development